Contact Information
Professional Summary
Experience
Education
Academic Interests
Awards
Skills
Languages
Projects

CV

Academic resume, research profile, honors, and technical background.

Contact Information

Name	Haoran Liu
Professional Title	Undergraduate Student in Computer Science and Technology
Email	liuhaoran@stu.neuq.edu.cn
Location	Qinhuangdao, Hebei
Website	https://HaoR724.github.io

Professional Summary

Undergraduate researcher at Northeastern University at Qinhuangdao with a GPA of 4.06/5.00 and a top-7% rank in the major. My work focuses on large language model evaluation, computer vision for micro-object detection, and planning-and-control methods for autonomous driving. I am actively seeking Master's and Ph.D. opportunities.

Experience

2024 - PRESENT
Qinhuangdao, China

Undergraduate Researcher

Northeastern University at Qinhuangdao

Conducting project-based research in LLM evaluation, UAV micro-object detection, and autonomous driving planning.

- Investigated gray-box benchmark contamination detection through option restructuring, log-probability analysis, and Isolation Forest.
- Developed the SR-TOD framework and DGFE module to improve tiny-object detection in complex maritime UAV imagery.
- Designed an integrated U-turn decision, path optimization, and MPC control pipeline for urban autonomous driving.

Education

2022 - 2026
Qinhuangdao, China

B.Eng.

Northeastern University at Qinhuangdao (NEUQ)

Computer Science and Technology

- Top 7% of the major with GPA 4.06/5.00.
- CET-6 score 555.
- Research interests in artificial intelligence, computer vision, and autonomous driving.

Academic Interests

Research Areas: Artificial Intelligence, Large Language Models, Computer Vision, Tiny Object Detection, Autonomous Driving

Awards

2025

University Comprehensive Third-Class Scholarship
Northeastern University at Qinhuangdao
Awarded in October 2025.

2025

University Comprehensive First-Class Scholarship
Northeastern University at Qinhuangdao
Awarded in April 2025.

2025

Innovation and Entrepreneurship Scholarship
Northeastern University at Qinhuangdao
Awarded in April 2025.

2024

University Comprehensive First-Class Scholarship
Northeastern University at Qinhuangdao

2024

National College Mathematical Modeling Contest, Provincial First Prize
China Undergraduate Mathematical Contest in Modeling

2024

National College Mathematical Modeling Contest, Provincial Second Prize
China Undergraduate Mathematical Contest in Modeling

Skills

Programming Languages: C/C++, Python, Java

AI and Computer Vision: PyTorch, OpenCV, Isolation Forest, Kalman Filter, Model Predictive Control

Tools and Utilities: MATLAB, LaTeX, Git

Languages

Chinese : Native speaker

English : CET-6 555; strong academic reading and technical writing ability

Projects

Gray-box Data Leakage Detection in LLM Benchmarks

Investigated the reliability of benchmark evaluation under opaque pre-training data conditions.

- Generated derivative benchmark sequences for MMLU and CMMLU through multiple-choice option restructuring.
- Queried target models such as LLaMA 2 and Qwen for log-probability distributions.

- Applied Isolation Forest to identify contaminated cases under deliberate option scrambling.

SR-TOD for Micro-Object Detection in UAV Imagery

Built a high-precision visual detection pipeline for tiny objects under waves, glare, and uneven illumination.

- Combined SuperGlue-based stitching with weighted fusion and Multi-Scale Retinex preprocessing.
- Proposed the Self-Reconstruction Tiny Object Detection framework with a lightweight reconstruction head.
- Designed the Difference-Guided Feature Enhancement module to improve multi-scale feature discrimination.

Autonomous Driving U-turn Decision and Path Optimization

Constructed a perception-prediction-decision-planning-control pipeline for complex urban U-turn scenarios.

- Predicted dynamic obstacles with Kalman Filter and quantified safety margins with Time-to-Collision.
- Planned smooth trajectories with cubic Bezier curves and Analytic Hierarchy Process.
- Used Model Predictive Control for low-risk trajectory refinement and high-risk emergency braking.

© Copyright 2026 Haoran Liu. Built with Jekyll and deployed on GitHub Pages. Last updated: April 11, 2026.